NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

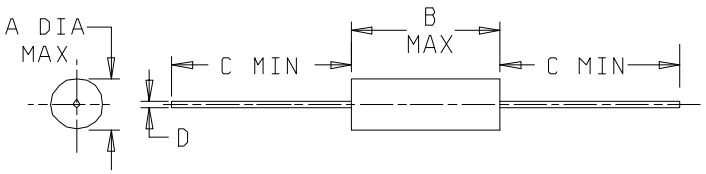
MIL-PRF-27/286B
5 April 1990
SUPERSEDING
MIL-T-27/286A
12 July 1982

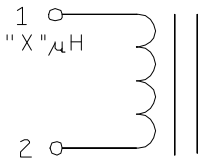
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE)
INDUCTORS, POWER, TF5R04ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.





WV: 50 VOLT PK
ALTITUDE: 70,000 FT MAX
(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

Dimensions	
A	.375 (9.53)
B	.500 (12.7)
C	1.25 (31.8)
D	.032 (0.81)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Marking shall be on the side of the case.
5. Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations.

(B) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:02Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Electrical ratings See table I

TABLE I Electrical ratings.

Dash number 1/	Inductance µH (1-2) +30-5% 2/ 3/	DC current amps (1-2) max 3/ 4/	DC resistance ohms (max) (1-2) 3/	Voltage (1-2) 3/
01	120	1.6	0.14	1 volt 20 kHz
02	300	1.0	0.35	1 volt 20 kHz
03	500	0.75	0.56	1 volt 20 kHz
04	1200	0.50	1.40	1 volt 20 kHz
05	3000	0.30	3.50	1 volt 20 kHz
06	5000	0.25	5.60	1 volt 20 kHz
07	30	3	.035	1 volt 20 kHz
08	50	2 5	.056	1 volt 20 kHz

1/ Qualification testing and approval to M27/286-06 shall be sufficient to grant qualification approval to M27/286-01 through M27/286-08.
2/ The inductance is measured with 0 A dc applied to (1-2) and at the specified voltage across (1-2).
3/ When numbers are in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.
4/ At rated dc current, inductance will exceed 80 percent of rated inductance.

Design and construction

Dimensions and configuration See figure 1.

Duty cycle Continuous.

Case Epoxy.

Terminals .032 inch oxygen free copper leads.

Weight 4.5 grams, maximum.

Altitude. 75,000 feet, maximum.

Operating temperature range -55°C to +105°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:02Z
Check the source to verify that this is the current version before use.

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage

At sea level: 100 volts rms.

At reduced barometric pressure: 100 volts rms

Vibration (high frequency) MIL-STD-202, method 204.

Temperature rise: 45°C with 1 volt rms, 20 kHz, current as specified in table I applied to (1-2) at an ambient temperature of 60°C maximum.

Marking location. See figure 1

Part number M27/286- (dash number from table I).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0744)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T04:02Z
Check the source to verify that this is the current version before use.